

EAST SALE, Australia

WMO: 949070

Lat: 38.1156S

Lon: 147.1322E

Elev: 8

StdP: 101.23

Time Zone: 10.00 (AUS)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	0.1	1.3	-0.8	3.5	5.5	0.1	3.8	5.7	13.7	12.1	12.3	12.1	2.0	310	0.533

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	11.7	33.1	19.5	30.0	19.2	27.3	18.6	21.4	28.4	20.5	26.5	19.7	25.0	5.0	310

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
19.6	14.3	22.8	18.7	13.5	22.2	17.8	12.8	21.6	62.2	28.5	59.0	26.9	56.2	25.0	25.5

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
12.2	10.7	9.4	DB	-2.5	39.4	1.1	2.2	-3.3	41.0	-4.0	42.2	-4.6	43.5	-5.4	45.1	
			WB	-2.3	23.5	1.2	1.1	-3.2	24.3	-3.9	24.9	-4.6	25.5	-5.5	26.3	

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	14.0	19.4	19.3	17.5	14.3	11.7	9.3	8.8	9.6	11.4	13.4	15.6	17.5
	DBStd	4.59	3.44	3.11	3.05	2.59	2.19	2.08	2.00	2.08	2.46	2.85	3.11	3.28
	HDD10.0	137	0	0	0	1	8	37	47	31	10	2	0	0
	HDD18.3	1779	25	22	54	123	207	271	294	269	207	156	94	57
	CDD10.0	1585	293	260	232	131	59	16	11	20	53	108	169	232
	CDD18.3	185	59	49	27	3	0	0	0	0	1	3	13	30
	CDH23.3	1831	548	386	247	48	3	0	0	0	14	85	183	316
	CDH26.7	730	251	153	90	11	0	0	0	0	2	23	67	133
(14) Wind	WSAvg	3.7	4.1	3.9	3.6	3.1	3.4	3.3	3.5	3.7	3.8	3.8	4.1	3.9
(15) Precipitation	PrecAvg	581	44	38	45	49	50	46	43	44	52	51	61	55
	PrecMax	886	125	166	154	137	190	195	167	112	205	129	140	153
	PrecMin	303	1	0	5	7	9	4	7	5	12	5	16	2
	PrecStd	136	29	34	32	33	43	38	31	20	28	26	34	36
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	38.2	35.6	33.5	27.7	22.7	18.6	18.4	21.0	25.3	29.7	32.5	35.8
		MCWB	21.1	20.5	19.2	16.9	14.4	12.6	11.5	12.9	14.7	17.2	18.7	19.7
	2%	DB	32.9	31.2	29.0	24.2	20.0	16.2	16.1	18.0	22.0	25.5	28.2	30.4
		MCWB	20.0	19.7	18.9	16.3	13.9	11.4	10.5	11.6	13.7	15.8	17.9	19.1
	5%	DB	29.0	27.7	25.8	21.5	18.0	14.9	14.7	16.2	19.2	22.5	24.8	26.6
		MCWB	19.8	19.7	18.3	15.4	13.0	11.0	10.1	10.8	13.0	14.9	16.9	18.4
	10%	DB	25.7	25.0	23.2	19.6	16.6	13.8	13.4	14.7	17.0	19.5	21.8	23.7
		MCWB	18.7	18.9	17.4	14.7	12.3	10.5	9.7	10.0	12.0	14.0	16.0	17.4
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	23.0	22.5	21.3	18.4	16.1	13.9	12.7	13.9	16.4	18.5	20.5	21.7
		MCDB	31.6	29.8	27.5	24.6	20.1	16.1	15.4	19.0	22.6	24.4	27.7	29.7
	2%	WB	21.5	21.2	20.0	17.2	14.7	12.6	11.5	12.5	14.8	16.8	18.8	20.2
		MCDB	29.0	27.3	25.1	21.5	18.3	15.1	14.3	16.3	19.4	23.1	25.4	26.7
	5%	WB	20.4	20.3	19.1	16.3	13.8	11.7	10.7	11.4	13.5	15.6	17.7	19.0
		MCDB	26.7	25.5	24.1	20.0	17.0	14.1	13.5	14.8	18.0	21.0	23.0	24.8
	10%	WB	19.4	19.4	18.1	15.4	12.9	10.9	10.1	10.6	12.4	14.5	16.6	18.0
		MCDB	24.8	23.9	22.5	18.7	15.8	13.2	13.0	13.8	16.3	19.0	20.9	22.8
(35) Mean Daily Temperature Range	5% DB	MDBR	11.7	11.2	11.3	11.1	10.2	9.7	9.6	10.3	11.2	11.6	11.3	11.6
		MCDBR	18.0	16.4	16.3	14.5	12.6	11.2	11.6	13.3	15.4	17.7	17.3	17.6
		MCWBR	6.7	6.6	6.9	7.3	7.4	7.2	7.1	7.8	8.5	8.7	7.6	7.3
	5% WB	MCDBR	14.8	13.2	13.2	12.2	11.7	10.3	10.1	11.7	13.6	15.1	14.7	14.6
		MCWBR	6.3	5.9	6.1	6.4	7.3	7.1	6.6	7.4	8.0	8.0	7.0	6.7
(40) Clear-Sky Solar Irradiance	taub		0.384	0.369	0.341	0.333	0.312	0.301	0.303	0.310	0.334	0.344	0.352	0.380
	taud		2.415	2.482	2.551	2.549	2.583	2.605	2.586	2.563	2.493	2.490	2.496	2.413
	Ebn at Noon		948	938	926	870	831	814	833	884	918	954	974	957
	Edh at Noon		123	110	95	83	69	63	68	80	98	108	112	124
(44) All-Sky Solar Radiation	RadAvg		6.83	5.87	4.80	3.41	2.35	1.87	2.13	2.96	4.14	5.38	6.22	6.92
	RadStd		0.43	0.37	0.31	0.21	0.12	0.13	0.11	0.13	0.30	0.36	0.40	0.41

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only		+0.33	N/A	N/A	N/A	+0.50	+0.69	N/A	-104	+137	+42
(47) Regional (28 neighbors)		+0.48	N/A	N/A	+0.75	N/A	N/A	-14	-128	+169	+59

Nomenclature: See separate page